Nguyen Manh Hiep

Software Engineer

0978618337 Hiep.work@outlook.com linkedin.com/in/hiepnm2003 nguyenmanhhiep.info

SUMMARY

I am a software engineer who builds full-stack products and puts AI and automation to work inside them. At ChimAI I delivered a notarial document platform with AI OCR and the backend, dashboards and infrastructure of a sales-automation SaaS; as a freelancer I built a farm operations system alone. I design REST APIs and the data model behind them, and care most about the parts that must stay correct: queues, schedulers, idempotent webhooks and offline sync. I ship with Docker, Nginx and GitHub Actions, and automate my own work with Claude Code and an assistant I run on OpenClaw.

EXPERIENCE

Notarex - notarial document digitization platform 03/2026 - 09/2026

Software Engineer, ChimAI JSC (Full-stack + AI document processing + Infrastructure):

- **Description:** A platform that digitizes notarial paperwork for notary offices in Vietnam, covering configurable form templates, AI-driven OCR extraction and a full document edit, save and download cycle.
- **Technologies & Tools:** Next.js 14, TypeScript, FastAPI, SQLAlchemy async, PostgreSQL 16, Celery, Redis, MinIO, OnlyOffice
- **Key Responsibilities & Contributions:**
  - **AI OCR Pipeline:** Generated each document type's extraction prompt from its template configuration, so a new document type needs configuration, not code; QR-coded national IDs skip the model.
  - **Document Workflow & Authorization:** Built the flow from uploaded photos through human review to a JWT-protected OnlyOffice edit-save-download cycle with docxtpl DOCX generation, under a three-role model scoped per notary office.
  - **Background Processing:** Built a Celery and Redis task queue on a ZSET pending queue with an atomic active-counter, plus heartbeat reconciliation that recovers zombie jobs, keeping OCR stable at scale.
  - **Requirements & Delivery:** Collected requirements directly with notary offices, scoped each release in sprints in a 2 to 3 person team with no dedicated PM, and shipped on Docker behind Nginx with MinIO storage and GitHub Actions CI/CD.

EazySale - sales lead generation and outreach SaaS 03/2026 - 08/2026

Software Engineer, ChimAI JSC (Backend + React dashboards + Infrastructure):

- **Description:** A SaaS for sales lead management and automated customer outreach over Zalo and email. I owned the FastAPI backend, three React dashboards and the infrastructure.
- **Technologies & Tools:** React 18, TypeScript, FastAPI, SQLAlchemy, Alembic, PostgreSQL 15, APScheduler, Zalo OA API, SendGrid, Docker
- **Key Responsibilities & Contributions:**
  - **Zalo OA Integration:** Built the Zalo OA ZNS API channel with template management, HMAC-SHA256 webhook verification and delivery metrics, plus a mock provider with a tunable failure rate for testing without spending quota.
  - **Scheduled Campaign Delivery:** Paired APScheduler date triggers with a 60-second database poll that recovers jobs lost to a restart, using SELECT FOR UPDATE NOWAIT so each message is sent exactly once.
  - **Email Event Tracking:** Made the public SendGrid event webhook idempotent with in-batch deduplication and a status-priority guard inside the UPDATE, so out-of-order and repeated events cannot roll a delivery state backwards.
  - **Architecture & Delivery:** Designed a path-routed Nginx deployment on one domain, RBAC across 129 REST endpoints, and a GitHub Actions pipeline deploying four images with migrations and zero downtime.

ASC Farm - farm production management system 06/2026 - 08/2026

Freelance Software Engineer, sole developer (Backend + Web admin + Mobile PWA):

- **Description:** A web admin and offline-capable mobile PWA that runs an intercropped farm from cultivation plans to verified field tasks.
- **Technologies & Tools:** Frappe v15 (headless), Python, MariaDB 11, Redis, React 18, TypeScript, Leaflet, PWA, Docker
- **Key Responsibilities & Contributions:**
  - **Task Generation Engine:** Generated field tasks from cultivation processes with prerequisite chains and "N times every N days" frequencies, idempotently, regenerating them when a process or crop cycle changes.

- **Workload Balancing:** Assigned each task to a team lead, preferring the block owner and otherwise the least-loaded lead, and reassigned open work automatically when a lead became inactive.
- **Offline-first Field App:** Built an idempotent offline queue keyed by a client UUID, a service-worker app shell, and in-app completion photos stamped with GPS and a watermark as proof of work.
- **Testing & Delivery:** Covered the engine, status model and permissions with 139 backend tests across 65 API endpoints, packaged as a one-command Docker Compose stack.

PROJECTS

SenAI - real-time multilingual meeting translation 2026 - present

Frontend, Backend and AI Integration Owner (Real-time systems + LLM orchestration + Infrastructure):

- **Description:** Live meeting translation by a 5-person team for VAIC 2026 (three awards), running at app.senai.online.
- **Technologies & Tools:** Next.js 16, TypeScript, FastAPI, PostgreSQL + pgvector, LiveKit, vLLM, Soniox, ElevenLabs
- **Key Responsibilities & Contributions:**
  - **AI/LLM Integration:** Built an extensible model router for ASR, translation, TTS and LLM that switches provider at runtime between self-hosted vLLM and commercial APIs, with fallback handling on failure.
  - **Translation Pipeline:** Ran an independent ASR and translation pipeline per microphone to separate speakers, pairing fast translation with LLM refinement against a contextual glossary.
  - **Meeting Intelligence:** Added Soniox transcription that separates three or more speakers, LLM-written minutes with an assistant that answers questions about them, and per-speaker voiced translation with ElevenLabs.
  - **Delivery & Security:** Deployed through GitHub Actions over SSH with a forced command that skips API restarts for doc-only commits, revoked refresh tokens on logout, and kept 37 UAT suites with 147 real, non-mocked tests.

Ely - personal AI assistant built on OpenClaw 2026

Extended the open-source OpenClaw agent with Telegram commands, CDP browser automation, cron jobs for daily chores, my own skills that dispatch and supervise Claude Code tasks, and a self-restarting gateway.

EDUCATION

Hanoi University of Science and Technology (HUST) 2021 - 2026 (expected)

Computer Science

HONORS & AWARDS

Third Prize (top 3 overall of 300+ teams), Vietnam AI Innovation Challenge 2026
Vietnam AI Stars Award, Vietnam AI Innovation Challenge 2026
AI Singapore Award, Vietnam AI Innovation Challenge 2026

TECHNICAL SKILLS

AI-assisted development	Claude Code (agents, skills, hooks), OpenClaw agents, MCP servers, Cursor, Copilot, prompt & context engineering, RAG, LLM orchestration, model routing with fallback, vLLM
Frontend	React 19/18, Next.js 16/14, TypeScript, Vite, Tailwind v4, Zustand, PWA / offline-first
Backend	FastAPI, Python, Frappe v15, REST API design, Node.js / NestJS, Spring Boot, WebSocket, LiveKit, Celery, Redis queues, JWT / OAuth, RBAC, webhooks
Databases	PostgreSQL (+ pgvector), MariaDB, MySQL, SQLite, SQLAlchemy 2.0 async, MinIO (S3)
DevOps & Cloud	Docker Compose, Nginx, systemd, TLS, VPS deployment, GitHub Actions CI/CD
Engineering practice	System design, SOLID, design patterns, code review, idempotency & concurrency control, workflow & scheduling engines, non-mocked integration testing, SRS / Use Case / ERD
Soft Skills	Critical thinking, technical research, cross-functional teamwork, fast adaptability